JULIAN CABRERA MARCEGLIA

Smart Manufacturing & Digital Transformation Engineer | Industry 4.0 | Manufacturing Intelligence | Industrial Data Platforms

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PROFESSIONAL SUMMARY

Manufacturing Systems professional with 21+ years of experience transforming manufacturing operations through the integration of industrial expertise, digital technologies and data-driven improvement initiatives.

Experienced in automotive manufacturing environments supporting Toyota production programs, with a career evolution from shop-floor operations into manufacturing quality systems, industrial data platforms and digital manufacturing solutions.

Specialized in identifying manufacturing transformation opportunities, analyzing operational challenges and designing digital solutions that connect shop-floor processes, manufacturing data and engineering decision-making.

Designed and implemented an Industrial Measurement Data Platform that transformed fragmented dimensional measurement information into a structured manufacturing data environment, integrating multiple measurement sources and centralizing more than 50,000 records.

Developed a Real-Time Quality Prioritization System applying manufacturing analytics concepts to transform quality data into prioritized operational insights and improvement actions.

Combines deep manufacturing process knowledge with software engineering and industrial data capabilities to support Smart Manufacturing initiatives, factory digitalization programs and operational excellence improvements.

CORE EXPERTISE

Smart Manufacturing & Digital Transformation

- Industry 4.0 Concepts
- Smart Manufacturing Solutions
- Factory Digitalization
- Digital Transformation Initiatives
- Manufacturing Intelligence
- Manufacturing Information Systems
- Data-Driven Manufacturing Improvement
- Manufacturing Process Optimization
- Operational Excellence Initiatives
- Continuous Improvement

Industrial Data & Analytics

- Industrial Data Platforms
- Manufacturing Data Integration
- Engineering Data Management
- Manufacturing Analytics
- Advanced Manufacturing Analytics
- Predictive Quality Analytics
- KPI Monitoring and Visualization
- Statistical Process Analysis
- Decision Support Systems

Manufacturing Transformation

- Shop-floor Digitalization
- Manufacturing Process Analysis
- Cross-functional Stakeholder Collaboration
- Engineering and Production Alignment
- Manufacturing Information Visibility
- Quality and Process Improvement

PROFESSIONAL EXPERIENCE

Metalsa

Sep 2016 – Present | Buenos Aires Province, Argentina

Quality Technician Supervisor

Support Toyota automotive manufacturing programs by combining manufacturing operations expertise, quality engineering knowledge and digital manufacturing capabilities.

Lead initiatives focused on improving manufacturing visibility, engineering decision-making and data-driven operational improvement.

Digital Manufacturing & Transformation Initiatives

- Designed and implemented an Industrial Measurement Data Platform integrating multiple manufacturing measurement sources into a centralized engineering information environment.
- Transformed fragmented manufacturing information into structured digital datasets supporting engineering analysis and operational decisions.
- Developed workflows for data integration, validation, processing and reporting.
- Centralized more than 50,000 manufacturing measurement records to improve traceability and analytical capability.
- Automated reporting workflows, reducing report generation effort approximately 10x.
- Improved accessibility of manufacturing information for quality and engineering teams.
- Established scalable foundations supporting future digital manufacturing initiatives.

Manufacturing Intelligence & Operational Excellence

- Developed a Real-Time Quality Prioritization System applying manufacturing analytics concepts to identify quality risks and prioritize engineering actions.
- Converted manufacturing measurement information into actionable insights supporting data-driven improvement decisions.
- Supported the transition from reactive quality monitoring toward predictive and proactive manufacturing improvement.
- Lead KPI review meetings and cross-functional improvement initiatives involving engineering, production and quality teams.
- Support root-cause investigations and continuous improvement activities through statistical and data-driven analysis.
- Perform SPC, MSA, GR&R and process capability studies.
- Provide technical authority for dimensional acceptance and rejection decisions.

Metalsa

Sep 2008 – Sep 2016

Quality Technician

Supported automotive manufacturing quality operations for Toyota programs.

- Generated and analyzed manufacturing information through inspection and measurement systems.
- Performed dimensional analysis, SPC studies and measurement-system evaluations.
- Developed and maintained hundreds of inspection programs.
- Supported engineering teams in identifying process improvement opportunities.
- Improved manufacturing process understanding through data analysis and quality-system expertise.

Metalsa

Sep 2004 – Sep 2008

Production Welder

Performed automotive manufacturing operations in production environments.

Developed practical understanding of shop-floor processes, manufacturing constraints and operational requirements that later supported digital manufacturing transformation initiatives.

SELECTED DIGITAL TRANSFORMATION PROJECTS

Industrial Measurement Data Platform

Manufacturing Data Infrastructure | Digital Factory Enablement

Designed and implemented a manufacturing data platform transforming fragmented dimensional measurement information into structured engineering intelligence.

Challenge: Manufacturing information was distributed across multiple measurement systems, creating challenges related to: Data accessibility, Traceability, Reporting efficiency, Engineering analysis.
Solution: Implemented: Multi-source measurement data integration, Structured engineering datasets, Automated reporting workflows, Manufacturing information management processes, Analytics-ready manufacturing data environment.
Impact: Centralized 50,000+ measurement records. Improved manufacturing information visibility. Reduced reporting effort approximately 10x. Enabled scalable support for increasing product complexity.

Real-Time Quality Prioritization System

Manufacturing Intelligence | Advanced Analytics | Operational Decision Support

Developed a manufacturing intelligence solution transforming dimensional measurement information into prioritized engineering actions.

Capabilities: Automated manufacturing data processing, Process capability analysis, Quality risk prioritization, Data-driven decision support, Engineering improvement indicators.
Impact: Improved identification of critical manufacturing risks. Reduced manual analysis effort. Supported proactive quality improvement initiatives. Established a scalable framework for manufacturing intelligence.

TECHNICAL CAPABILITIES

Digital Manufacturing

- Industry 4.0
- Smart Manufacturing
- Factory Digitalization
- Manufacturing Intelligence
- Manufacturing Data Platforms
- Manufacturing Systems Integration
- Digital Transformation

Industrial Data & Analytics

- Python
- SQL
- DuckDB
- Pandas
- NumPy
- SciPy
- Statistical Process Control
- Capability Analysis
- Manufacturing Analytics
- Data Visualization

Manufacturing Engineering

- Automotive Manufacturing
- Production Processes
- Dimensional Engineering
- CMM Systems
- FARO Measurement Systems
- PC-DMIS
- GD&T
- Quality Systems

PROFESSIONAL POSITIONING

Manufacturing Systems professional combining hands-on manufacturing experience, industrial data engineering capabilities and digital transformation expertise.

Proven ability to translate manufacturing challenges into scalable digital solutions that improve operational visibility, engineering decision-making and continuous improvement.

Focused on enabling Smart Manufacturing environments by connecting shop-floor operations, manufacturing processes, data and technology.